

Monte Carlo Simulations of $D = 4$ Matrix Model

Project Report



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Abstract

In this project we study the bosonic $D = 4$ matrix model on a lattice with the help of the Hamiltonian Monte Carlo (HMC) algorithm. We have studied various observables of the model such as the Polyakov loop, extent of space, and internal energy.

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Contents

1	Introduction	5
2	Hamiltonian Monte Carlo	5
2.1	Acceptance Rate	6
2.2	Auto-correlation	6
2.3	Error Estimation	6
3	Lattice Formulation of the $D = 4$ Model	6
3.1	The $D = 4$ Model	7
3.2	HMC for $D = 4$ Model	10
3.3	Observables	11
4	Simulation Details	12
5	Results and Conclusion	13

1 Introduction

The bosonic $D = 4$ model is an important model to study with the help of lattice simulations. It is an introductory step to study the more famous BFSS (black hole thermodynamics) and IKKT matrix models (spacetime compactification). Here we have used the Hamiltonian Monte Carlo (HMC) algorithm to study this model. To study the behavior of this model at various temperatures, we have used the Polyakov loop as the order parameter.

2 Hamiltonian Monte Carlo

The classic 1953 paper of Metropolis et al. introduced to us the world of Markov chain Monte Carlo (MCMC). Unlike Monte Carlo integration, the goal of MCMC is not to sample a multi-dimensional region uniformly. Instead, the goal is to visit a point x with a probability proportional to some given distribution function say, $p(x)$. The Markov process of generating one state after other is referred to as updating. In this process, we change the initial state randomly by an arbitrarily small quantity. Since the acceptance of an updated state depends on the ratio of the distribution functions at the initial and updated states, field configurations closer to the solution state have a greater probability of being accepted. The starting point of the simulations can be arbitrary, and it can be far away from the equilibrium values. However, the value must lie within the configuration space. For example, if we know that the solution is complex, it is better not to start with real numbers as thermalization would take far too long or may not take place at all. Thermalization happens when the observable starts fluctuating around a fixed value over a significant number of sweeps [1]. If we give a fixed field configuration as a starting point then it is known as the cold start.

HMC is an example of Markov chain Monte Carlo (MCMC). Suppose we wish to sample from a probability distribution. We begin by constructing a Hamiltonian in terms of this probability distribution. On top of the variables we are interested in (they are the position variables in HMC language), we should also introduce some auxiliary field variables (momentum variables in HMC). Variables typically would have independent distributions that are Gaussian. The HMC method draws these momentum variables from a Gaussian distribution and then computes a trajectory according to a discretized version of Hamiltonian dynamics. At the end of the trajectory the new proposed state is accepted or rejected by a Metropolis step. The advantage of this method is that it can propose a new state that is far away from the current state, with a high probability of acceptance. This would be a huge gain compared to the random walk Metropolis.

2.1 Acceptance Rate

Since we are using the Hamiltonian Metropolis algorithm in our simulations, the best results are obtained if we keep our acceptance rate between 80% and 95%. This is because of the precise nature of the algorithm, as too low an acceptance would result in the acceptance ultimately dropping off to zero, if the field configuration deviates significantly from the optimum value, while much higher acceptance is avoided as the action takes much longer to thermalize.

2.2 Auto-correlation

The auto-correlation function is used to determine the correlation between neighboring field configurations in the simulation. We have

$$C_X(X_i, X_{i+t}) = \langle (X_i - \bar{X}_i)(X_{i+t} - \bar{X}_{i+t}) \rangle = \langle X_i X_{i+t} \rangle \langle X_i \rangle \langle X_{i+t} \rangle. \quad (1)$$

It would have been unnecessary in case if the successive field configurations were chosen arbitrarily, but as it is not the case, we have to determine the auto-correlation time, or the period after which the auto-correlation between them is negligible. This is because configurations separated by this period are nearly independent of each other, and hence can be used in determining the expectation value of a particular observable using known statistical formulas like

$$\bar{X} = \frac{\sum_{i \leq N} X_i}{N}, \quad \sigma_X^2 = \frac{\sum_{i \leq N} \langle X_i - \bar{X} \rangle^2}{N}. \quad (2)$$

Thus due to auto-correlation the effective number of observations that are used to compute the expectation values of the various observables will be less than those produced in the simulations.

2.3 Error Estimation

The main reason for using auto-correlation time is that since the observables that are separated by the autocorrelation time are uncorrelated, that is, independent, hence, if we only consider such values while determining the expectation value, we can use the mean square formula for calculating the error.

3 Lattice Formulation of the $D = 4$ Model

In order to perform numerical simulations, we must discretize time, at the end of which the temporal co-ordinate takes on the form of a periodic lattice. In this section

we try to describe how to implement gauge fields on a lattice by using an example of $D = 4$ model, whose action is obtained by dimensionally reducing the $3 + 1$ -dimensional Yang-Mills model with gauge group $SU(N)$ to $0 + 1$ dimensions, by dimensional reduction, described in detail in Ref. [2].

The lattice formulation is a necessary prerequisite for using Monte Carlo simulations to analyze such actions that are hard to study otherwise.

3.1 The $D = 4$ Model

The Euclidean action of the $D = 4$ model is given by

$$S_E = \frac{N}{\lambda} \int_0^\beta dt \operatorname{Tr} \left[\frac{1}{2} (D_t X_i)^2 - \frac{1}{4} [X_i, X_j]^2 \right], \quad (3)$$

where $D_t X^i = \partial_t X^i - i[A(t), X^i]$ is the covariant derivative. As always repeated indices are summed over, $i, j = 1, 2, \dots, d$, $\lambda = g^2 N$, where λ is the 't Hooft coupling and g is the one-dimensional Yang-Mills coupling, X_i are $N \times N$ traceless Hermitian matrices, and $A(t)$ is the gauge field.

We also have periodic boundary conditions for both the scalar and gauge fields, i.e., $X^i(t + \beta) = X^i(t)$ and $A(t + \beta) = A(t)$. For this model, $d = 3$ since we have three scalar fields for the $D = 4$ model.

The gauge transformations are

$$X^i(t) \rightarrow V(t) X^i(t) V^\dagger(t), \quad (4)$$

$$A(t) \rightarrow V(t) (A(t) + i \partial_t) V^\dagger(t), \quad (5)$$

where $V(t) \in SU(N)$. The partition function is given by

$$Z = \int \mathcal{D}A \mathcal{D}X \exp\{-S_E\}. \quad (6)$$

The whole procedure of putting the bosonic BFSS model (which has an identical action as our model) on a lattice is described in Refs. [3] and [2]. The same arguments are used here as well. First, we note that here we only have temporal direction and so we will have a one-dimensional lattice and that lattice is given by $\Lambda = \{n | n = 0, 1, \dots, T - 1; t = na\}$ and imposing boundary conditions implies that we need to identify 0 with T and $\beta = aT$. If we discretize the ordinary derivative we get

$$\partial_t X_n^i \rightarrow \frac{X_{n+1}^i - X_n^i}{a}. \quad (7)$$

To discretize the covariant derivate we need to introduce the link fields given by

$$U_{n,n+1} = \mathcal{P} \left\{ \exp \left[\imath \int_{na}^{(n+1)a} dt A(t) \right] \right\} = \exp(\imath a A(t)), \quad (8)$$

where $\mathcal{P}$ indicates the path-ordered product. To make the above expression gauge covariant we have to transport back the field at t_{n+1} to t_n . For the discrete version of the covariant derivative, we obtain

$$\mathcal{D}_t X_n^i \rightarrow \frac{U_{n,n+1} X_{n+1}^i U_{n,n+1}^\dagger - X_n^i}{a}. \quad (9)$$

Substituting this back in the discretized action we get

$$S_E = \frac{Na}{2\lambda} \sum_{n=0}^{T-1} \text{Tr} \left\{ \left(\frac{U_{n,n+1} X_{n+1}^i U_{n,n+1}^\dagger - X_n^i}{a} \right)^2 - \frac{1}{2} [X_n^i, X_n^j]^2 \right\}. \quad (10)$$

Simplifying the above equation we get

$$S_E = \frac{N}{\lambda} \sum_{n=0}^{T-1} \text{Tr} \left\{ -\frac{U_{n,n+1} X_{n+1}^i U_{n,n+1}^\dagger X_n^i}{a} + \frac{(X_n^i)^2}{a} - \frac{a}{4} [X_n^i, X_n^j]^2 \right\}. \quad (11)$$

Now we use the $SU(N)$ symmetry to simply the action further. If we change the fields by the following set of transformations

$$\begin{aligned} X_0^i &\rightarrow X_0^i, \\ X_1^i &\rightarrow U_{0,1}^\dagger X_1^i U_{0,1}, \\ &\vdots \\ X_{T-1}^i &\rightarrow (U_{0,1} U_{1,2} \dots U_{T-2,T-1})^\dagger X_{T-1}^i (U_{0,1} U_{1,2} \dots U_{T-2,T-1}), \end{aligned} \quad (12)$$

and also if we define

$$W = (U_{0,1} U_{1,2} \dots U_{T-2,T-1}, U_{T-1,0}), \quad (13)$$

we can express the action in the following way

$$\begin{aligned} S_E &= -\frac{N}{a\lambda} \text{Tr} \left\{ W X_0^i W^\dagger X_{T-1}^i + \sum_{n=0}^{T-2} X_{n+1}^i X_n^i \right\} \\ &\quad + \frac{N}{\lambda} \sum_{n=0}^{T-1} \text{Tr} \left\{ \frac{(X_n^i)^2}{a} - \frac{a}{4} [X_n^i, X_n^j]^2 \right\}. \end{aligned} \quad (14)$$

Upon using the result that every (finite-dimensional) Hermitian matrix can be diagonalized, we can diagonalize the W matrix by using the gauge symmetry of the field X_n^i . Let W be diagonalized by V matrix then

$$W = VDV^\dagger, \quad (15)$$

where

$$D = \text{diag}(e^{i\theta_1}, e^{i\theta_2}, \dots, e^{i\theta_N}). \quad (16)$$

If we transform $X_n^i \rightarrow VX_n^iV^\dagger$ then the action becomes

$$S_E = -\frac{N}{a\lambda} \text{Tr} \left\{ DX_0^i D^\dagger X_{T-1}^i + \sum_{n=0}^{T-2} X_{n+1}^i X_n^i \right\} + \frac{N}{\lambda} \sum_{n=0}^{T-1} \text{Tr} \left\{ \frac{(X_n^i)^2}{a} - \frac{a}{4} [X_n^i, X_n^j]^2 \right\}. \quad (17)$$

We will use this action for the simulation purpose. The reason we did this was to simplify the integration measure of the system. We will see that the link fields would get considerably simplified due to this choice of link fields we have taken. The integration measure for the link fields is given by

$$\prod_{n=0}^{T-1} \mathcal{D}U_{n,n+1} = \prod_{n=1}^{T-1} \mathcal{D}U_{n,n+1} \mathcal{D}U_{0,1} = \prod_{n=1}^{T-1} \mathcal{D}U_{n,n+1} \mathcal{D}W. \quad (18)$$

The above relation is obtained because a defining property of the Haar Measure, as explained in Ref. [5], is that being a product measure, it is cyclic and using the relation $U_{0,1} = W(U_{1,2} \dots U_{T-2,T-1})^\dagger$ we can get the above relation.

The action depends only on the diagonal matrix D , which is the diagonalized form of W . So we can integrate the extra variables out and the integration measure

simplifies to

$$\begin{aligned}
Z &= \int \prod_{n=0}^{T-1} \mathcal{D}U_{n,n+1} \mathcal{D}X e^{-S[X,D]} \\
&= \int \prod_{n=1}^{T-1} \mathcal{D}U_{n,n+1} \mathcal{D}W \mathcal{D}X e^{-S[X,D]} \\
&= (\text{Vol}(SU(N)))^{T-1} \int \mathcal{D}W \mathcal{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^N d\theta_i \prod_{j>k}^N |e^{i\theta_j} - e^{i\theta_k}|^2 \mathcal{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^N d\theta_i \prod_{j>k}^N \sin^2 \left(\frac{\theta_j - \theta_k}{2} \right) \mathcal{D}X e^{-S[X,D]} \\
&\propto \int \prod_{i=1}^N d\theta_i \mathcal{D}X e^{-S[X,D(\theta)] - S_{FP}[\theta]}.
\end{aligned} \tag{19}$$

The proportionality constant does not bother us because it cancels out from the numerator and denominator when we calculate the expectation values of the observables. The Fadeev-Popov term in this equation is introduced due to the change of basis when we diagonalized W . We have

$$S_{FP}[\theta] = -2 \sum_{j<k}^N \ln \left| \sin \left(\frac{\theta_j - \theta_k}{2} \right) \right| = - \sum_{j \neq k}^N \ln \left| \sin \left(\frac{\theta_j - \theta_k}{2} \right) \right|. \tag{20}$$

Since $D \in SU(N)$, all the θ variables will not be independent and is constrained by the equation

$$\sum_{i=1}^N \theta_i = 0 \tag{21}$$

as $\det(D) = 1$.

3.2 HMC for $D = 4$ Model

Now we will introduce the conjugate momenta for every field variables. Momentum conjugate to X_n^i is P_n^i , where P_n^i is an $N \times N$ traceless Hermitian matrix and the momentum conjugate to θ_k is P_k .

The Hamiltonian for the molecular dynamics part of the HMC is

$$H = \sum_{n=0}^{T-1} \frac{1}{2} \text{Tr} \left\{ P_n^2 \right\} + \sum_{j=1}^N \frac{1}{2} P_j^2 + S[X, D(\theta)] + S_{FP}[\theta] \quad (22)$$

The Hamilton's equations are given by

$$\begin{aligned} \dot{X}_{n,rs}^i &= \frac{\partial H}{\partial P_{n,rs}^i} = P_{n,rs}^i & \dot{P}_{n,rs}^i &= -\frac{\partial H}{\partial X_{n,rs}^i}, \\ \dot{\theta}_j &= \frac{\partial H}{\partial P_j} = P_j & \dot{P}_j &= -\frac{\partial H}{\partial \theta_j}, \end{aligned}$$

where the dot represents time derivative with respect to the fictitious time τ . These forces can be calculated using the Hamiltonian, and are given by

$$-\frac{\partial H}{\partial X_{0,rs}^i} = \frac{N}{a\lambda} (X_1^i - 2X_0^i + D^\dagger X_{T-1}^i D)_{rs} + \frac{Na}{\lambda} [X_0^j, [X_0^i, X_0^j]]_{rs}, \quad (23)$$

$$\begin{aligned} -\frac{\partial H}{\partial X_{n,rs}^i} &= \frac{N}{a\lambda} (X_{n+1}^i - 2X_n^i + X_{n-1}^i)_{rs} + \frac{Na}{\lambda} [X_n^j, [X_n^i, X_n^j]]_{rs}, \\ &\text{for } n = 1, 2, \dots, T-2, \end{aligned} \quad (24)$$

$$\begin{aligned} -\frac{\partial H}{\partial X_{T-1,rs}^i} &= \frac{N}{a\lambda} (D^\dagger X_0^i D - 2X_{T-1}^i + X_{T-2}^i)_{rs} \\ &\quad + \frac{Na}{\lambda} [X_{T-1}^j, [X_{T-1}^i, X_{T-1}^j]]_{rs}, \end{aligned} \quad (25)$$

$$\begin{aligned} -\frac{\partial H}{\partial \theta^j} &= \frac{2N}{a\lambda} \sum_{k=1}^N \text{Re} \left(i X_{T-1,kj}^i X_{0,jk}^i e^{i(\theta_j - \theta_k)} \right)_{rs} \\ &\quad + \sum_{k=1, k \neq j}^N \cot \left(\frac{\theta_j - \theta_k}{2} \right). \end{aligned} \quad (26)$$

3.3 Observables

The observables we are interested in calculating are the Polyakov loop, the internal energy, and the extent of space for the bosonic $D = 4$ model. Here we give the definitions of each of them, in continuous as well as in discretized form.

1. **Polyakov loop** $|P|$: Polyakov loop is a modified Wilson loop, which is defined as any closed loop in spacetime coordinates. Due to the periodic boundary conditions in the $D = 4$ model, we cannot gauge transform all temporal links to identity. We can, however, gauge the spatial pieces of our loop to identity.

Then the Wilson loop reduces to the two disconnected paths of temporal link variables, located in space. Both these paths wind around the temporal direction of the lattice but have opposite orientations. We can make this new observable gauge-invariant by taking the trace for each of the two loops individually. It is also the order parameter for the confined / deconfined phase transition seen in the model. (See Ref. [5].) We have

$$\langle |P| \rangle = \frac{1}{N} \left\langle \left| P \left(e^{i \int dt A(t)} \right) \right| \right\rangle = \left\langle \left| \frac{\text{Tr}(D)}{N} \right| \right\rangle \quad (\text{on the lattice}). \quad (27)$$

2. **Extent of space R^2 :** The extent of space can be regarded as the gauge invariant version of region of the configuration space covered by the bosonic fields

$$\langle R^2 \rangle = \frac{\lambda}{N\beta} \left\langle \int_0^\beta dt \text{Tr}(X^{i^2}) \right\rangle = \frac{\lambda}{NT} \left\langle \sum_{n=0}^{T-1} dt \text{Tr}(X_n^{i^2}) \right\rangle \quad (\text{on the lattice}). \quad (28)$$

3. **Internal energy E :** Internal energy is the total energy of the system and is used for determining the order of the phase transition. We have

$$\begin{aligned} \left\langle \frac{E}{N^2} \right\rangle &= -\frac{3\lambda}{4N\beta} \left\langle \int_0^\beta dt \text{Tr} \left([X^i, X^j]^2 \right) \right\rangle \\ &= -\frac{3\lambda}{4NT} \left\langle \sum_{n=0}^{T-1} dt \text{Tr} \left([X_n^i, X_n^j]^2 \right) \right\rangle \quad (\text{on the lattice}). \end{aligned} \quad (29)$$

4 Simulation Details

In our simulations we changed the lattice spacing a to change the temperature while keeping the lattice size T fixed. Also, we opted for a large lattice size deliberately to keep the lattice spacing small, as large values of a would greatly increase the time taken for simulation.

We fixed the following values for every simulation: $d = 3$, $N = 4$, $\lambda = 1.0$, and $T = 26$. The total no of sweeps made were 500,000 and the thermalization was done till 100,000 sweeps. Since we are dealing with a large number of sweeps, and the computation took quite a while, the code was written in **bash** script to ensure that the field configurations were saved after every 200 sweeps. To make the computation faster, we ran the simulations for different temperatures in different

cores simultaneously. The temperature varies from 0.4 to 3.0 and while plotting, on retrospection, we increased the number of points near the transition region in order to get a much accurate value for the temperature T_c at which the phase transition takes place.

We fitted the Polyakov loop vs temperature plot using the equation

$$|P(T)| = A \arctan(B(T - T_c)) + D \quad (30)$$

and the fit values obtained are listed in Tab. 1. Here, T_c indicates the critical temperature at which the confined/deconfined phase transition takes place.

In the simulations, each variable and parameter has been measured in units of the 't Hooft coupling λ_0 . Since the Yang-Mills coupling g^2 has a dimension of $[L^{-3}]$ in one dimension, so X_in and $A(n)$ can be measured in the unit of $\lambda_0^{1/3}$ and can be measured in units of $\lambda_0^{-1/3}$. Thus, every variable and parameter appearing in the discrete action are dimensionless.

Fitting Constants	Fitted value
A	0.365907(11030)
B	1.22405(6783)
T_c	1.01857(20880)
D	0.511837(7843)

Table 1: Table showing the fitted values for the Polyakov loop vs temperature.

5 Results and Conclusion

The results are close to that of Refs. [4] and [2]. We can see that there is a visible phase transition from the plot of the Polyakov loop as a function of temperature. However, it is not a very sharp transition as the number of lattice sites are less (since even though the action we used is discretized, a large number of lattice sites means that we are effectively using a continuous action) and the errors are not small enough as the number of Monte Carlo sweeps are not very large, and due to autocorrelation, the effective number of Monte Carlo sweeps used for calculating the expectation values of the observables became very less indeed.

The Internal energy and the extent of space seem to increase almost linearly with temperature after reaching the critical temperature. However, it is important to note

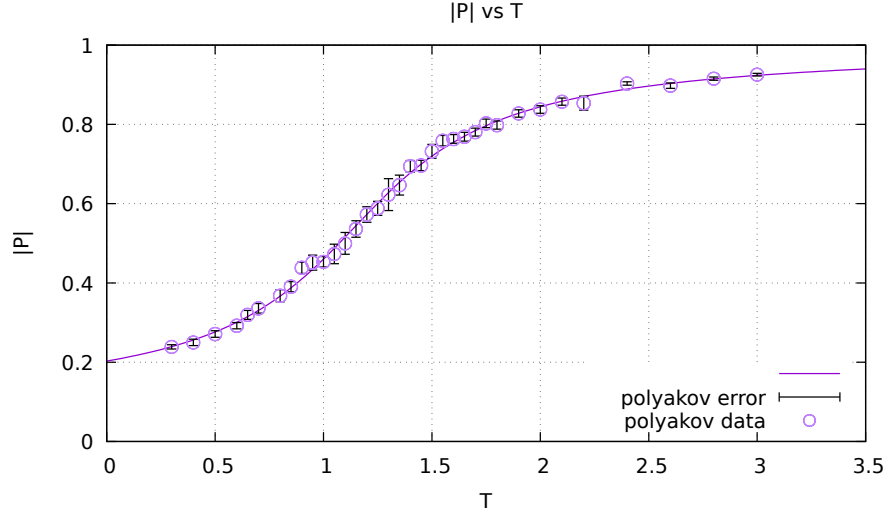


Figure 1: Polyakov loop vs temperature.

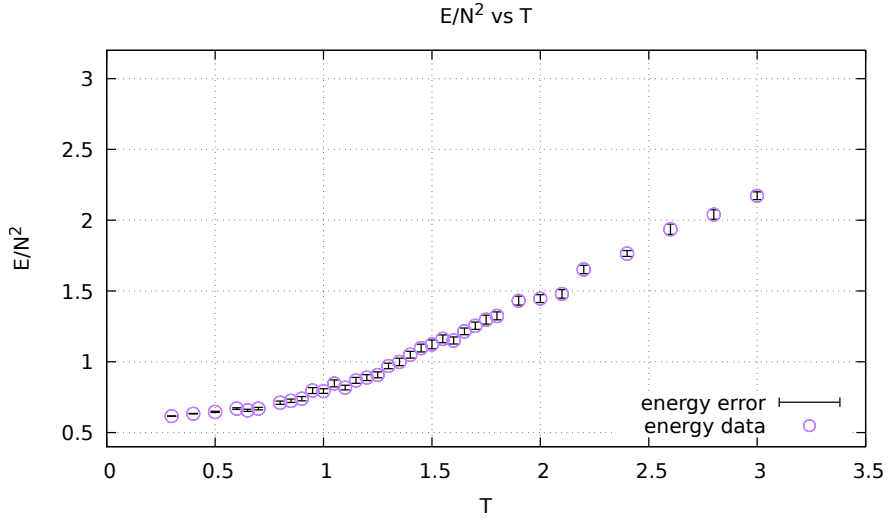


Figure 2: Internal energy vs temperature.

that the extent of space does not increase very rapidly, which would mean that field configurations are not stable (something which is known as flat directions).

However, the linear increase of the extent of space can be explained as the internal energy of the system also increases, and as the energy of the particles increase, they

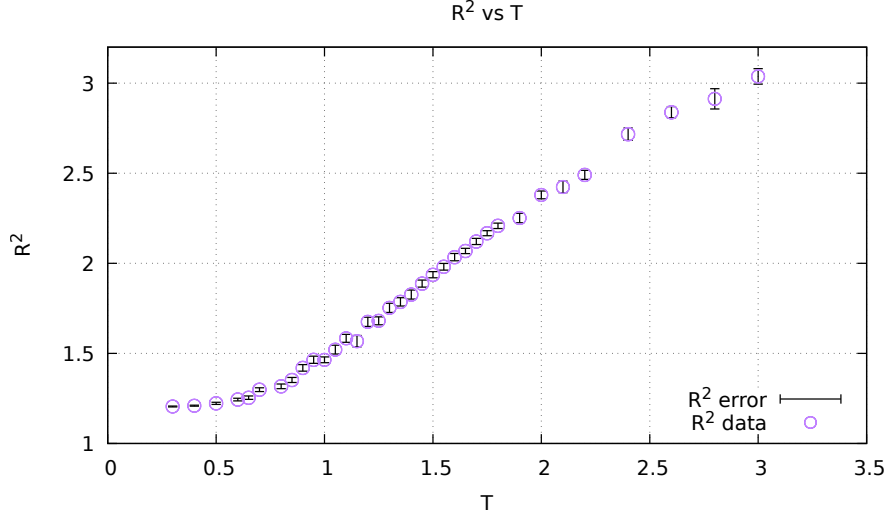


Figure 3: Extent of space vs temperature.

have access to greater regions in the configuration space, and hence should have greater extent of space.

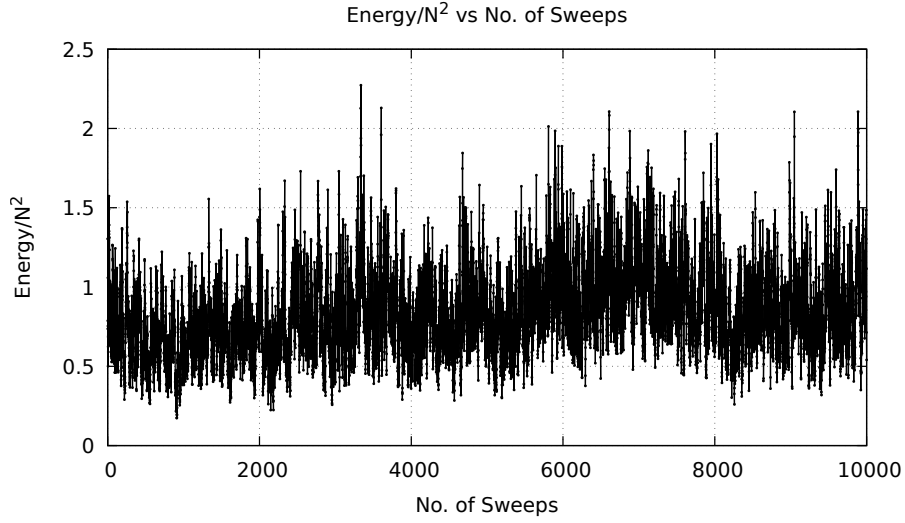


Figure 4: Energy vs number of sweeps (N) after thermalization ($N = 150,000$ to $160,000$, $T = 1.0$).

The above figures show the various observables after thermalization. One can

see that they oscillate about a fixed value. However, in both the Polyakov loop and the extent of space seems to oscillate about the the fixed value in almost a periodic manner, rather than random oscillations, which would be a solid line about the fixed value. Although, we have depicted only the graphs of a single temperature, they are representative of all graphs in the region of the phase transition.

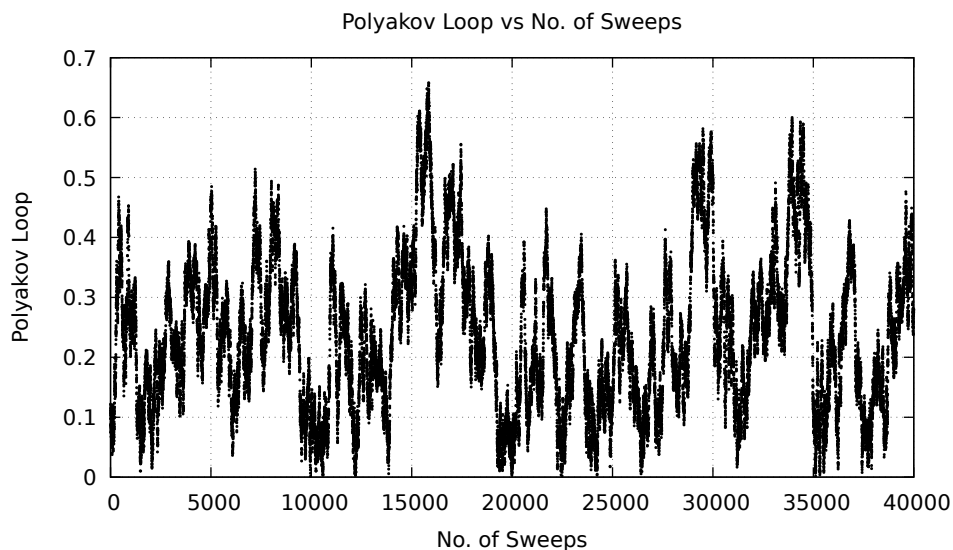


Figure 5: Polyakov loop vs number of sweeps (N) after thermalization ($N = 150,000$ to $190,000$ and $T = 1.0$).

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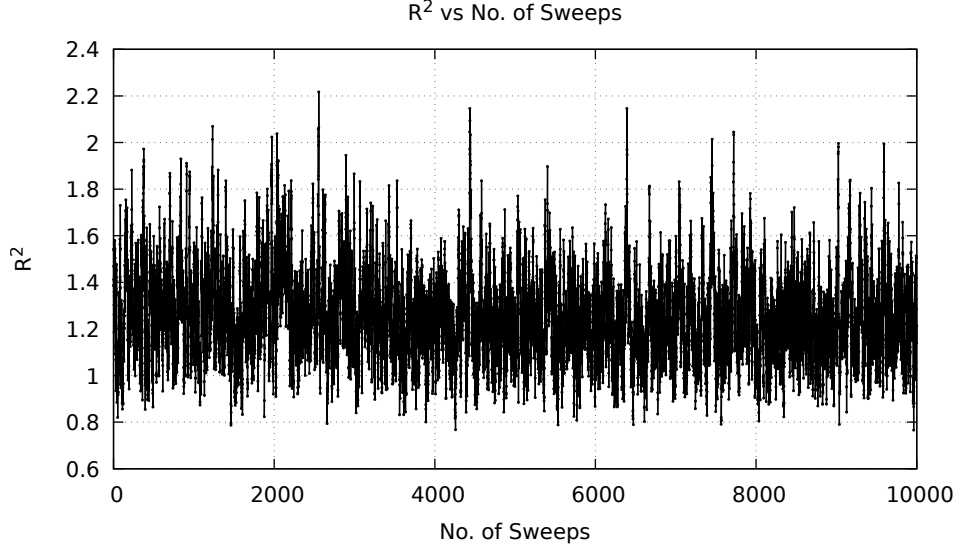


Figure 6: Extent of space vs number of sweeps (N) after thermalization ($N = 150,000$ to $160,000$ and $T = 1.0$).

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